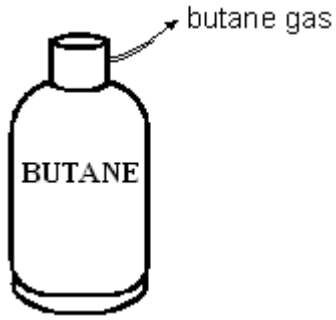
Q1.

The drawing shows a cylinder of butane. The butane is under pressure in the cylinder. Most of the butane in the cylinder is liquid, but some is a gas.



(a) Tick the **two** correct statements about the molecules of butane gas in the cylinder.

The molecules of gas are:

closer together than those in the liquid; ☐

further apart than those in the liquid; ☐

the same distance apart as those in the liquid; ☐

bigger than those in the liquid; ☐

smaller than those in the liquid; ☐

the same size as those in the liquid; ☐

2 marks

(b) The gas molecules exert a pressure on the inside of the cylinder.
How do the moving molecules cause this pressure?

.....

.....

1 mark

(c) (i) Explain why the pressure inside the cylinder falls when the temperature falls.

.....

.....

.....

.....

2 marks

(ii) Butane from the cylinder was used in a camping stove. If there was a large fall in the air temperature around the cylinder, what effect would this have on the flame of the camping stove?

.....

.....

1 mark
Maximum 6 marks

Q2.

Sue pumps up a bicycle tyre. As she does so, she notices that the pump becomes hot.

(a) Where, and how, was the energy stored before it was transferred in pumping up the tyre?

.....

1 mark

(b) Explain how the gas molecules inside the tyre exert pressure on the walls of the tyre.

.....

.....

1 mark

(c) The air going into the tyre was warmed up by the pumping. What effect will this have on the motion of gas molecules in the air in the tyre?

.....

.....

1 mark

(d) When the air in the tyre becomes hotter, the pressure rises. Give **one** reason, in terms of the motion of gas molecules in air, why the pressure rises.

.....

.....

1 mark

(e) The pressure in the tyre increases as Sue forces more air into the tyre. Explain why a larger number of gas molecules increases the pressure in the tyre.

.....

.....

1 mark
Maximum 5 marks

Q1.

(a) further apart than those in the liquid ✓

1 (L7)

the same size as those in the liquid ✓

1 (L7)

if more than two boxes are ticked, deduct one mark for each incorrectly ticked box
minimum mark zero

(b) by hitting **or** colliding with the inside of the cylinder

1 (L7)

(c) (i) particles move more slowly **or** have less energy

accept 'some of the particles condense into a liquid'
*do **not** accept 'particles move less **or** vibrate more slowly'*

1 (L7)

particles hit the inside of the cylinder less hard **or** less often

*do **not** accept 'do not press so hard'*

1 (L7)

(ii) lower **or** smaller **or** less hot

*accept 'dies out' or 'drops' **or** 'less strong' **or** 'goes out'*

1 (L7)

[6]

Q2.

(a) as chemical energy in Sue **or** her muscles

***both** parts are required for the mark*
*accept 'as chemical energy in glucose **or** named chemicals'*

1 (L6)

(b) the molecules collide with the walls of the tyre

*accept 'they hit the walls **or** the tyre'*
***or** 'they bounce off the walls'*

1 (L7)

(c) they speed up **or** get faster

1 (L7)

(d) any **one** from

- they will hit the tyre wall more frequently
- they hit the tyre wall harder **or** faster

accept 'more collisions with the tyre'
*do **not** accept 'more collisions'*
*do **not** accept 'the molecules move faster'*

1 (L7)

(e) there will be more frequent collisions with the tyre wall

accept 'more collisions with the tyre'
*do **not** accept 'more collisions'*
accept 'the force applied by the molecules increases'
*do **not** accept 'less space for the molecules'*
*do **not** accept 'there are more air molecules to hit the tyre wall'*

1 (L7)

[5]